

Session 2 · Deck 2.A2

Aiyagari: Production and the Capital Market

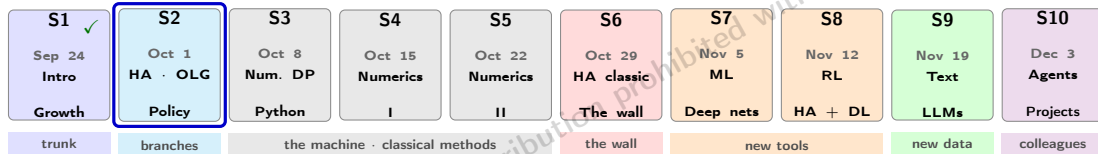
Zhigang Feng

Thursday, October 1, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

▼ you are here: 2.A2

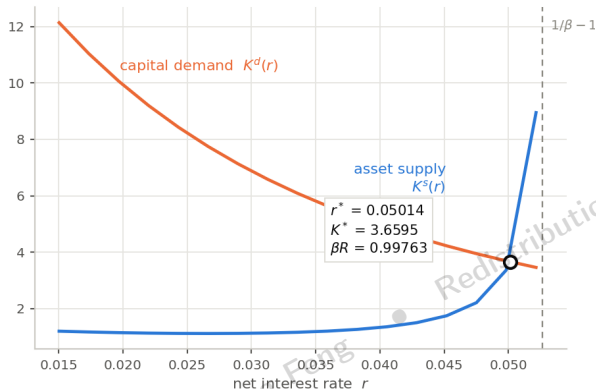


Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 2.A1 why insurance fails · 2.A2 Aiyagari: production and the capital market · 2.A3 the stationary distribution, and where the model is taken next.

The Picture from Week 1, Now with a Model Behind It

(a) Stationary equilibrium — bisection on r



A classical Aiyagari solve: capital supply from the household distribution against capital demand from the firm. We reproduce this figure ourselves in S6.

What changes from Huggett

The asset is no longer an IOU in zero net supply. It is a claim on **physical capital**, and its return is set by a technology.

- Households still face uninsurable risk and a borrowing limit.
- Now a firm stands ready to absorb savings at a return $r = F_K - \delta$.
- Equilibrium is a **crossing**, not a zero-sum condition.

The Environment

- A continuum of **ex-ante identical**, infinitely lived households, unit mass. Preferences $\mathbb{E}_0 \sum_t \beta^t u(c_t)$.
- **Idiosyncratic labour productivity** $z_t \in \mathcal{Z}$, a finite Markov chain with transition $\pi(z, z')$ and invariant distribution π^* . Labour is supplied inelastically, so

$$L = \sum_z z \pi^*(z)$$

is a constant. **No aggregate shock.**

- One asset: capital, held as a , paying $1 + r$. Borrowing limit $a' \geq -\underline{a}$.
- A representative firm with constant returns:

$$Y = F(K, L) = AK^\alpha L^{1-\alpha}, \quad \text{depreciation } \delta.$$

Ex ante vs. ex post

Households are identical *before* the shocks and different *after* them. All inequality in this model is the accumulated history of luck.

A law of large numbers is doing real work: individual risk averages out across the continuum, so aggregates are deterministic even though each household's life is not.

The Household, Recursively

Taking the constant prices (r, w) as given,

$$v(a, z) = \max_{c, a'} \left\{ u(c) + \beta \sum_{z'} v(a', z') \pi(z, z') \right\}$$

subject to

$$c + a' = (1 + r)a + wz, \quad a' \geq -\underline{a}, \quad c \geq 0.$$

What we solve for: the value $v(a, z)$, the asset policy

$a'(a, z)$, and consumption

$$c(a, z) = (1 + r)a + wz - a'(a, z).$$

The individual state is (a, z) — exactly the row we wrote for Aiyagari in the template in 1.B1.

This is an MDP

MDP	Aiyagari
state	(a, z)
action	a'
reward	$u(c)$
discount	β
transition	$\pi(z, z') \times \text{asset map}$

But it is **not** a standard MDP: the reward depends on (r, w) , which are themselves equilibrium objects. The environment is endogenous — the point of 1.B3.

The Firm Closes the Model

Constant returns means firm size is indeterminate, so we may speak of a single representative firm.

Competition in factor markets gives

$$r = \alpha A \left(\frac{K}{L} \right)^{\alpha-1} - \delta, \quad w = (1 - \alpha) A \left(\frac{K}{L} \right)^{\alpha}.$$

- L is **exogenous** — labour is inelastic and the shock distribution is invariant.
- So w is a function of r alone: pin down r and everything else follows.
- K is **endogenous** — it is whatever households have chosen to hold.

One unknown price

Two markets must clear, capital and labour, but labour clears trivially.

So the entire equilibrium reduces to finding **one number**: the interest rate r^* .

That is the strategy for the rest of the deck — and, in S6, the strategy for the code: parameterize everything by r , then find the r that clears the capital market.

Stationary Recursive Competitive Equilibrium (1/2)

Definition, part 1: prices and behaviour

A **stationary recursive competitive equilibrium** consists of factor prices (r^*, w^*) , a value function v , policies $(c(a, z), a'(a, z))$, a measure λ^* on $\mathcal{A} \times \mathcal{Z}$, and aggregates (K^*, L) such that:

1. **Firm optimality** — prices are marginal products:

$$r^* = \alpha A (K^*/L)^{\alpha-1} - \delta, \quad w^* = (1 - \alpha) A (K^*/L)^{\alpha}.$$

2. **Household optimality:** given (r^*, w^*) , the policies $c(a, z)$ and $a'(a, z)$ solve the household's problem, and v is the associated value function.

Market clearing and the stationarity of λ^* are on the next slide.

Stationary Recursive Competitive Equilibrium (2/2)

Definition, part 2: consistency

3. **Labour market clears:** $L = \sum_z \int z d\lambda^*(a, z).$
4. **Capital market clears:** $K^* = \sum_z \int a'(a, z) d\lambda^*(a, z).$
5. **The distribution reproduces itself:** for every Borel set $\mathcal{B}$ and every z' ,

$$\lambda^*(\mathcal{B}, z') = \sum_z \pi(z, z') \int_{\{a : a'(a, z) \in \mathcal{B}\}} d\lambda^*(a, z).$$

Conditions 4 and 5 are **joint**. The distribution determines K^* ; K^* determines prices; prices determine the policy; the policy determines the distribution.

Nothing here says λ^* exists, or is unique. That is 2.A3.

Capital Demand: the Firm's Side

Inverting the firm's first-order condition gives capital demand as a function of the interest rate:

$$K^d(r) = L \left(\frac{r + \delta}{\alpha A} \right)^{\frac{1}{\alpha-1}}.$$

- Strictly **decreasing** and continuous in r (since $\alpha < 1$).
- $K^d(r) \rightarrow \infty$ as $r \rightarrow -\delta$: capital is free, so demand is unbounded.
- $K^d(r) \rightarrow 0$ as $r \rightarrow \infty$.

This is the ordinary neoclassical demand schedule. Nothing about heterogeneity enters it.

All the novelty is on the other side

The firm does not care *who* owns the capital, only how much there is.

Heterogeneity enters entirely through the **supply** curve — which is an integral over a distribution we have to compute.

Capital Supply: the Distribution's Side

Fix a candidate r . Solve the household problem, obtain $a'(a, z; r)$, let λ_r be the induced stationary distribution, and define

$$K^s(r) = \sum_z \int a \, d\lambda_r(a, z).$$

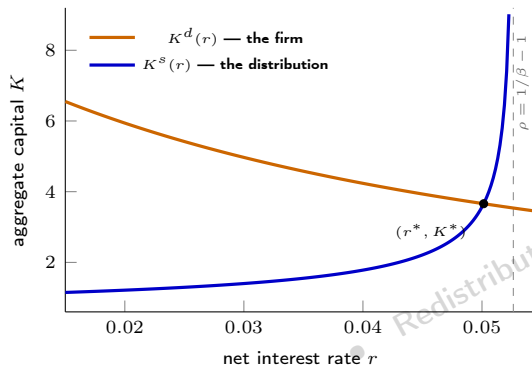
- **Complete markets.** Every household sets $\beta(1+r) = 1$, so desired assets are infinite at $r = \rho \equiv 1/\beta - 1$ and the supply curve has a **vertical asymptote** there.
- **Incomplete markets.** Precautionary saving makes $K^s(r)$ finite and positive at interest rates strictly *below* ρ — the curve shifts right.

An expensive function

Each point on $K^s(r)$ costs one full household solve plus one stationary-distribution computation.

This is why Aiyagari is a *nested* fixed point, and why it takes us until S6 to compute it honestly.

The Equilibrium Is Where They Cross



Same axes as the computed figure three slides back. Demand exact from $r = \alpha A (K/L)^{\alpha-1} - \delta$ at the Stage-1A calibration $\alpha = 0.34$, $\delta = 0.06$, $A = 1$, $L = 0.6633$, which puts the crossing at $r^* = 0.0501$, $K^* = 3.66$; supply schematic, asymptotic to $\rho = 1/\beta - 1$ with $\beta = 0.95$.

- Equilibrium: $K^d(r^*) = K^s(r^*)$.
- Because K^s is finite below ρ , the crossing occurs at

$$r^* < \rho$$

and therefore at a capital stock **above** the complete-markets level.

Over-accumulation. Uninsurable risk makes the economy hold more capital than it would with full insurance — and that extra capital is *not* a free lunch. It is the cost of self-insurance.

Why the Curves Must Cross

Two boundary conditions, then one theorem.

- $K^d(r)$ is continuous and strictly decreasing, with $K^d(r) \rightarrow \infty$ as $r \rightarrow -\delta$ and $K^d(r) \rightarrow 0$ as $r \rightarrow \infty$.
- $K^s(r)$ is bounded below by $-\underline{a}$, and $K^s(r) \rightarrow \infty$ as $r \uparrow \rho$.

So excess demand $K^d - K^s$ is **positive** near $r = -\delta$ and **negative** near $r = \rho$. If it is continuous, the intermediate value theorem puts a crossing in $(-\delta, \rho)$ — and the second boundary condition is what forces it to sit at $r^* < \rho$.

Everything now rests on the word *if*.

What continuity costs

$K^s(r) = \sum_z \int a d\lambda_r$ is continuous only if

- $a'(a, z; r)$ is continuous in r — the maximum theorem;
- λ_r **exists**, is **unique**, and moves continuously with r , which *presupposes* the first two.

Acemoglu (2009), Cor. 6.1; Stokey–Lucas–Prescott (1989), Thm. 12.13. This is 2.A3.

And a crossing is not *the* crossing. The sign of $\partial K^s / \partial r$ is ambiguous, so the curves may meet more than once.

What the Model Is Used For

- **Wealth and income inequality** — Gini, top shares, mobility.
- **Social insurance and tax-transfer policy** — who gains, who loses, and by how much.
- **Public debt** — crowding out versus supplying a safe asset households want.
- **Asset pricing** with idiosyncratic risk.
- **Entrepreneurship** and occupational choice.

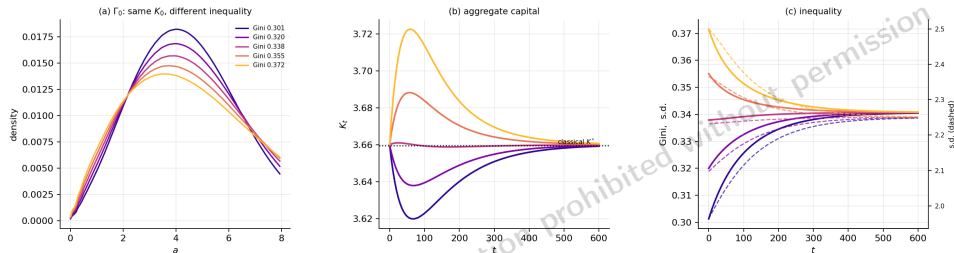
Research in Practice

Every model in the frictions pillar of 1.A1 is this engine with an extra state: health status, job-based insurance, business ownership.

Krueger, Mitman & Perri (2016, *Handbook*); Kaplan, Moll & Violante (2018, *AER*).

Every one of these is a claim about a **distribution**. The next slide shows that the distribution is a moving object — not merely a fixed point we solve for once.

Research in Practice: Every Start Ends in the Same Place



Five starts with the **same** aggregate capital $K_0 = 3.660$ and different inequality (Gini 0.301–0.372); the dotted line in (b) is the classical $K^* = 3.6595$.

Terminal K spreads by 3.5×10^{-8} across the five, and the largest L_1 distance between terminal densities is 6.0×10^{-7} . $T = 2000$, plotted to $t = 600$.

Research in Practice

Feng, Han, Sargent and Zhu (2026), work in progress. A stationary solver returns the **fixed point**; train one on the whole map $\Gamma_{t+1} = \Phi(\Gamma_t)$ and you get the **dynamical system** — convergence becomes a *result*, not an assumption. Read (a) with (c): every start has the **same** aggregate K_0 , yet the cross-sections travel for hundreds of periods. A first-moment summary of the state would see *nothing happening here* — 2.B1's problem.

Takeaways

- Aiyagari = Huggett + a neoclassical firm. The asset becomes capital, a wage appears, and the equilibrium becomes a **crossing** of capital demand and capital supply.
- Everything reduces to **one price**: labour clears trivially, w is a function of r , and the whole model is the search for r^* .
- Capital supply is an **integral over a stationary distribution**. It is the expensive object, and the reason this model needs a computer.
- Precautionary saving puts the crossing at $r^* < \rho$, so the economy **over-accumulates** capital relative to complete markets.
- The crossing exists because demand and supply run off in opposite directions at the two ends of $(-\delta, \rho)$ — but only if K^s is **continuous**, which is a claim about the distribution, not about the firm.

Bridge: What Exactly Is λ^* ?

We have used λ^* freely — to define market clearing, to integrate capital supply, to draw a curve. We have now *named* what the crossing assumes about it; we have not established any of it.

- Does an invariant distribution **exist**?
- Is it **unique**? If not, which one does your simulation find?
- Does iterating from an arbitrary starting distribution **converge** to it?
- Is $K^s(r)$ even **continuous** in r — without which the crossing argument has no force?

Next (2.A3)

Three properties of one operator — **Feller, mixing, monotonicity** — answer all four questions, and each has a two-state counterexample showing what goes wrong without it.

Then: what this model gets right about inequality, what it misses, and the extensions built to close the gap.